

CONTACTS

AFFILIATION: Politecnico di Milano
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RESEARCH INTERESTS

- Trustworthy scientific ML
- Generative modeling
- Differential geometry
- Geometric deep learning
- Information geometry
- Physics-aware architectures
- Unsupervised learning
- Signal processing

EDUCATION

2020 - 2025 Ph.D. in Information Technology
Politecnico di Milano, Milan, Italy
THESIS: “Machine Learning Methods for Signal Processing
with Applications to the Wireless Physical Layer” [\[link\]](#)
ADVISOR: Prof. Matteo Matteucci
CO-ADVISOR: Dr. Dario Tagliaferri

2017 - 2020 M.Sc. in Computer Science and Engineering (AI track)
Politecnico di Milano, Milan, Italy
THESIS: “Deep Learning-Based Low-Rank Space-Time Subspace
Prediction for mmWave MIMO Channels”
ADVISOR: Prof. Matteo Matteucci
CO-ADVISORS: Prof. Umberto Spagnolini, Dr. Dario Tagliaferri
FINAL GRADE: 110/110 *cum laude* (GPA: 28.95/30)

2014 - 2017 B.Sc. in Information Technology Engineering
Università del Salento, Lecce, Italy
THESIS: “WebRTC-based volatile peer-to-peer network for anonymization
purposes in confidential cloud services supply”
ADVISOR: Prof. Ing. Mario A. Bochicchio
FINAL GRADE: 110/110 *cum laude* (GPA: 29.95/30)

RESEARCH VISITS

Sep 2025 - Nov 2025 Visiting Postdoctoral Researcher
Department of Applied Mathematics and Computer Science
Danmarks Tekniske Universitet (DTU), Kgs. Lyngby, Denmark
SUPERVISOR: Prof. Søren Hauberg

AIM: Design a generative model with proper uncertainty quantification
for Grassmann manifold-valued data exploiting the information
geometry of probability distributions on the Grassmannian.

Sep 2023 - Oct 2023 Visiting PhD student
Department of Applied Mathematics and Computer Science
Danmarks Tekniske Universitet (DTU), Kgs. Lyngby, Denmark
SUPERVISOR: Prof. Søren Hauberg

AIM: Study learned Riemannian metrics in the latent space of generative models and apply them to the design of machine learning-based methods for effective wireless channel estimation.

ACADEMIC POSITIONS

Dec 2023 - ongoing Research fellow (Assegnista di ricerca L. 240/2010)
Artificial Intelligence and Robotics Lab (AIRLab)
Politecnico di Milano, Milan, Italy
RESEARCH PROJECT: Development of an electromagnetic digital twin of an urban vehicular setting in the context of 6G communications.

Sep 2020 - Oct 2020 Research fellow (Assegnista di ricerca L. 240/2010)
Artificial Intelligence and Robotics Lab (AIRLab)
Politecnico di Milano, Milan, Italy
RESEARCH PROJECT: Machine learning for mmWave channel estimation in vehicular scenarios.

RESEARCH PROJECTS

Nov 2023 - Mar 2025 **EM Digital Twin for V2X Communications**

Working package within the "Joint Communication and Sensing and its applications" research project, developed within the *Huawei Technologies Italia/Politecnico di Milano Joint Lab*.
AIM: Design a state-of-the-art electromagnetic (EM) digital twin simulation framework to faithfully model the radio-propagation environment in urban vehicular communications taking into account the impact of different moving vehicle meshes, and assess the performance of machine learning-based positioning algorithms on data generated from the devised accurate simulation environment.

Sep 2021 - May 2023 **Joint Communication and Sensing at the Infrastructure (JC&S@I)**

Research project within the *Huawei Technologies Italia/Politecnico di Milano Joint Lab*.
AIM: Design and evaluate machine learning-based solutions for joint radar communications at the infrastructure to improve the performance of communication systems with respect to existing standards.

Nov 2019 - Jan 2021 **Evidence-based MIMO (EBMIMO)**

Research project within the *Huawei Technologies Italia/Politecnico di Milano Joint Lab*.

AIM: Design and evaluate deep learning-based and unsupervised learning-based channel estimation methods for improving standard channel estimation techniques in mmWave/sub-THz Vehicle-to-everything (V2X) scenarios.

PUBLICATIONS IN CONFERENCE PROCEEDINGS

- **Cazzella, L.**, Hauberg, S., Arvanitidis, G., Matteucci, M. (2026). On the Latent Information Geometry of the Grassmann Manifold, 2026. In the Proceedings of the Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS 2026). Contributions: conceptualization, methodology, experiments, writing, review.
- Xu, Z., Mura, S., Linsalata, F., **Cazzella, L.**, Badini, D., Spagnolini, U. (2025). Enhancing 5G-based Localization in Dynamic Environments through Network Digital Twins. Published in the Proceeding of the IEEE Global Communications Conference (IEEE GLOBECOM) 2025. Contributions: methodology, writing, review.
- **Cazzella, L.**, Linsalata, F., Maleki, M., Badini, D., Matteucci, M., Spagnolini, U. (2025). Chartwin: a Case Study on Channel Charting-aided Localization in Dynamic Digital Network Twins. Published in 2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall). Contributions: conceptualization, methodology, experiments, writing, review.
- Zhu, M., Mura, S., Linsalata, F., **Cazzella, L.**, Badini, D., Spagnolini, U. (2025). AI-empowered Real-Time Line-of-Sight identification via network digital twins. Published in the Proceedings of IEEE INFOCOM 2025-IEEE Conference on Computer Communications Workshops. Contributions: methodology, review.
- **Cazzella, L.**, Linsalata, F., Magarini, M., Matteucci, M., Spagnolini, U. (2024). A multi-modal simulation framework to enable digital twin-based V2X communications in dynamic environments. Presented at the 2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall), Washington D.C., USA. Contributions: conceptualization, methodology, experiments, writing, review.
- Spano, C., Badini, D., **Cazzella, L.**, Matteucci, M. (2024, September). Digital Pre-Distortion with Deep Reinforcement Learning for 5G Power Amplifiers. Published in the Proceedings of the 2024 54th European Microwave Conference (EuMC) (pp. 1050-1053). Contributions: methodology, writing, review.
- Zhu, M., **Cazzella, L.**, Linsalata, F., Magarini, M., Matteucci, M., Spagnolini, U. (2024). On the Digitization of the EM Environment: A Comparison of Ray Launching Solutions. Published in the Proceedings of 2024 IEEE International Mediterranean Conference on Communications and Networking (MeditCom) (pp. 85-90). Contributions: methodology, writing, review.
- Lomurno, E., Archetti, A., **Cazzella, L.**, Samele, S., Di Perna, L., Matteucci, M. (2022, September). SGDE: Secure generative data exchange for cross-silo federated learning. Published in the Proceedings of the 2022 5th international conference on artificial intelligence and pattern recognition (pp. 205-214). Contributions: methodology, experiments, writing, review.
- **Cazzella, L.**, Tagliaferri, D., Mizmizi, M., Matteucci, M., Badini, D., Mazzucco, C., Spagnolini, U. (2022). Position-agnostic algebraic estimation of 6G V2X MIMO channels via unsupervised learning. Presented at the 2022 IEEE Wireless Communications and

Networking Conference (WCNC), Austin (TX), USA. Contributions: conceptualization, methodology, experiments, writing, review.

PUBLICATIONS IN INTERNATIONAL JOURNALS

- Del Moro L., Linsalata F., **Cazzella L.**, Spagnolini U., Magarini M. (2026). Robust Channel Estimation via Uncertainty-Aware Environmental Digital Twins. Pre-print. Under submission to the IEEE Transactions on Communications. Contributions: conceptualization, writing, review.
- Spano, C., Badini, D., **Cazzella, L.**, Matteucci, M. (2025). Local and Remote Digital Pre-Distortion for 5G Power Amplifiers with Safe Deep Reinforcement Learning. Published in Sensors, 25(19), 6102. Contributions: methodology, writing, review.
- Paglierani, N., Linsalata, F., Sevimay, O. A., **Cazzella, L.**, Badini, D., Magarini, M., Spagnolini, U. (2025). Digital network twin-enabled synchronization and localization. Published in IEEE Journal on Selected Areas in Communications (IEEE JSAC). Contributions: methodology, writing, review.
- **Cazzella, L.**, Linsalata, F., Badini, D., Matteucci, M., Magarini, M., Spagnolini, U. (2025). High-Fidelity RF Mapping: Assessing Environmental Modeling in 6G Network Digital Twins. Published in Computer Networks, Elsevier. Contributions: conceptualization, methodology, experiments, writing, review.
- Zhu, M., Linsalata, F., Mura, S., **Cazzella, L.**, Badini, D., Spagnolini, U. (2025). Exploiting Age of Information in Network Digital Twins for AI-driven Real-Time Link Blockage Detection. Published in Computer Networks, Elsevier. Contributions: methodology, writing, review.
- Zhu, M., **Cazzella, L.**, Linsalata, F., Magarini, M., Matteucci, M., Spagnolini, U. (2024). Toward real-time digital twins of EM environments: Computational benchmark for ray launching software. Published in the IEEE Open Journal of the Communications Society, 5, 6291-6302. Contributions: methodology, writing, review.
- **Cazzella, L.**, Mizmizi, M., Tagliaferri, D., Badini, D., Matteucci, M., Spagnolini, U. (2024). Deep learning-based target-to-user association in integrated sensing and communication systems. Published in the IEEE Journal of Selected Topics in Signal Processing, 18(5), 886-900. Contributions: conceptualization, methodology, experiments, writing, review.
- **Cazzella, L.**, Tagliaferri, D., Mizmizi, M., Badini, D., Mazzucco, C., Matteucci, M., Spagnolini, U. (2022). Deep learning of transferable MIMO channel modes for 6G V2X communications. Published in IEEE Transactions on Antennas and Propagation, 70(6), 4127-4139. Contributions: conceptualization, methodology, experiments, writing, review.

TEACHING AND TUTORING

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| Mar 2026 | Ray Tracing and AI for the Physical Layer (guest lecture)
PhD in Information Technology; Politecnico di Milano
Seminar held as part of the course “Communication technologies for autonomous and connected systems”. |
| Jan 2023 - Feb 2023 | AI and signal processing (course; 16h)
Master in AI & ML; Cefriel S.c.a.r.l. and Politecnico di Milano |

LECTURING/EXERCISE SESSIONS:

1. Deep learning methods for channel estimation.
2. Neural networks for power amplifiers dynamic modeling.
3. Neural network-aided self-interference cancellation schemes.
4. Complexity reduction methods for DL-based signal processing.

Nov 2022 - Jul 2023 **System and methods of contrast to nonlinear distortions based on neural networks for high-capacity radio links** (tutoring; 30h)

Master in AI & ML; Cefriel S.c.a.r.l. and Politecnico di Milano

AIM OF THE PROJECT: Study and development of neural network models for power amplifier modeling and digital pre-distortion, in line with the main themes of my research on the design of machine learning-driven methods for signal processing.

GRANTS AND ACKNOWLEDGEMENTS

- Jul 2026 **CINECA ISCRA-C grant**; award: 8.500 GPU/h on the Leonardo Cluster. PI: Dr. Lorenzo Cazzella; joint participation with Prof. Matteo Matteucci and Dr. Dario Tagliaferri. Grant awarded by the CINECA (Italian inter-university consortium for supercomputing).
- Dec 2025 **Huawei Tech Arena 2025 - 3rd prize**; award value: €1000; joint participation with Dr. Francesco Linsalata with a proposal on “AI-based RF and Antenna Design”. Grant awarded by Huawei Technologies to the top-3 teams out of more than 100 participants from both Italian and European universities.
- Sep 2025 **NVIDIA Academic Grant**; award: 4 RTX PRO 6000 Blackwell Max-Q GPUs; joint participation with Prof. Matteo Matteucci and Dr. Francesco Linsalata. Grant awarded by NVIDIA to support the project “Physics-Informed AI-based Channel Simulation for V2X Communications”.
- Jun 2025 **CINECA ISCRA-C grant**; award: 11.124 GPU/h on the Leonardo Cluster. PI: Dr. Lorenzo Cazzella; joint participation with Prof. Matteo Matteucci and Dr. Dario Tagliaferri. Grant awarded by the CINECA (Italian inter-university consortium for supercomputing).
- Oct 2024 **Featured article in a Special Article Collection** on *Antennas and Propagation for Emerging 5G/6G Communications* [[link](#)] of IEEE Transactions on Antennas and Propagation (75th anniversary of the Antennas and Propagation Society) for the manuscript “Deep learning of transferable MIMO channel modes for 6G V2X communications.” [[webpage link](#); [YouTube video link](#)].
- Jul 2024 **CINECA ISCRA-C grant**; award: 10.000 GPU/h on the Leonardo Cluster. Participation as group member; PI: Prof. Matteo Matteucci. Grant awarded by the CINECA (Italian inter-university consortium for supercomputing).
- Apr 2022 **IEEE WCNC 2022 ComSoc Travel Grant**; award value: \$650. Grant awarded by the IEEE Communication Society for participating to IEEE WCNC 2022, held in Austin (TX), USA.
- Sep 2020 **Three-year doctoral scholarship (open subject)**. Granted from the Italian Ministry of Instruction, University and Research.

SUPERVISED M.Sc. THESES

- Feb 2025 - ongoing **AI for Cooperative Perception in Vehicular Radar Networks**
(co-supervised with Dr. Dario Tagliaferri)
AIM: Design a generative deep learning architecture to perform physically-consistent super-resolution tasks on multi-static radar images.
- Sep 2023 - Dec 2024 **Geometric Characterization and Segmentation of High-Frequency Campus Environments for Enhanced Ray-Tracing Channel Simulations** (co-supervised with Dr. Francesco Linsalata)
AIM: Build and segment geometric features in high-frequency campus environments, aiming to enhance the performance of ray-tracing-based channel simulations.
- Sep 2023 - Dec 2024 **Geometrically Enriched Latent Spaces for Subspace Inference**
(co-supervised with Prof. M. Matteucci)
AIM: Model latent-variable generative models for subspace prediction by means of geometrically enriched variational autoencoder architectures.
- Feb 2023 - Dec 2023 **Vehicle-to-Vehicle Cooperative LiDAR Perception Based on Graph Neural Networks** (co-supervised with Prof. M. Matteucci)
AIM: Develop a deep learning method for the exchange of LiDAR point clouds to improve the extent and quality of the perceived environment at each ego-vehicle in vehicular cooperative perception.
- Feb 2022 - Dec 2022 **Camera-aided Dataset Generation for Radar Object Detection Models** (co-supervised with Prof. M. Matteucci and Dr. S. Mentasti)
AIM: Evaluate deep learning-based detection algorithms on radar targets over experimentally acquired indoor and outdoor radar images.

LANGUAGES

ITALIAN : Native speaker
ENGLISH : Fluent
GERMAN : Intermediate
FRENCH : Intermediate
DANISH : Elementary